



CS3001

COMPUTER NETWORKS

FINAL PROJECT

REPORT

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Roll No: 23i-2089

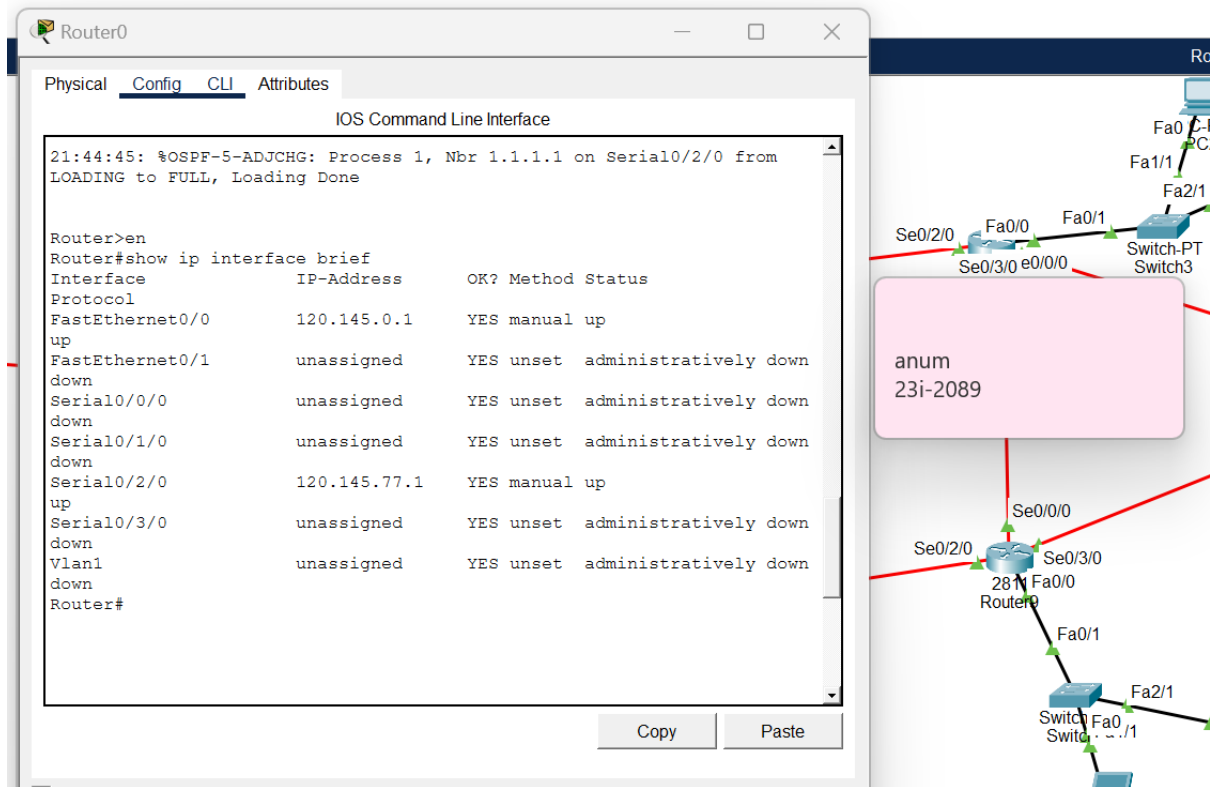
Submission Date: 7th December, 2025

1. Basic IP / Routing per Block

1.1. Show router IPs & masks (VLSM proof)

By **show ip interface brief** on Router0, Router1 and Router8, usage of /24 and /27 and /30 is showed.

On router0:



The image shows a screenshot of the Router0 CLI window and a network diagram. The CLI window displays the output of the `show ip interface brief` command, showing the status of various interfaces. The network diagram illustrates the physical connections between Router0, Switch-PT, and Switch3, including interfaces like Fa0/0, Fa0/1, Fa1/1, Fa2/1, Se0/2/0, Se0/3/0, and Se0/0/0.

Router0 CLI Output:

```
Router>en
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status
FastEthernet0/0	120.145.0.1	YES	manual	up
FastEthernet0/1	unassigned	YES	unset	administratively down
Serial0/0/0	unassigned	YES	unset	administratively down
Serial0/1/0	unassigned	YES	unset	administratively down
Serial0/2/0	120.145.77.1	YES	manual	up
Serial0/3/0	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down

Network Diagram:

The diagram shows a network topology with the following components and connections:

- Router0:** Connected to Switch-PT via Se0/2/0 (Fa0/0) and Se0/3/0 (e0/0/0).
- Switch-PT:** Connected to Router0 and Switch3 via Fa0/1 and Fa2/1.
- Switch3:** Connected to Switch-PT via Fa1/1 and Fa2/1.
- Router8:** Connected to Switch3 via Fa0/0 and Fa2/1.

A pink box labeled "anum 23i-2089" is overlaid on the diagram, pointing to the connection between Router0 and Switch-PT.

On router1:

Router1

Physical Config CLI Attributes

IOS Command Line Interface

LOADING to FULL, Loading Done

21:44:35: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/2/0 from LOADING to FULL, Loading Done

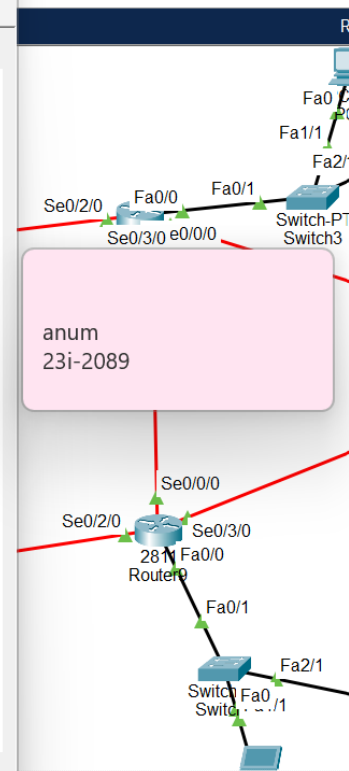
```
Router>en
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status
FastEthernet0/0	120.145.128.1	YES	manual	up
FastEthernet0/1	120.146.0.1	YES	manual	up
Serial0/0/0	120.145.77.5	YES	manual	up
Serial0/1/0	unassigned	YES	unset	administratively down
Serial0/2/0	120.145.77.9	YES	manual	up
Serial0/3/0	120.145.77.2	YES	manual	up
Vlan1	unassigned	YES	unset	administratively down

Router#

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On Router8

Router8

Physical Config CLI Attributes

IOS Command Line Interface

up: new adjacency

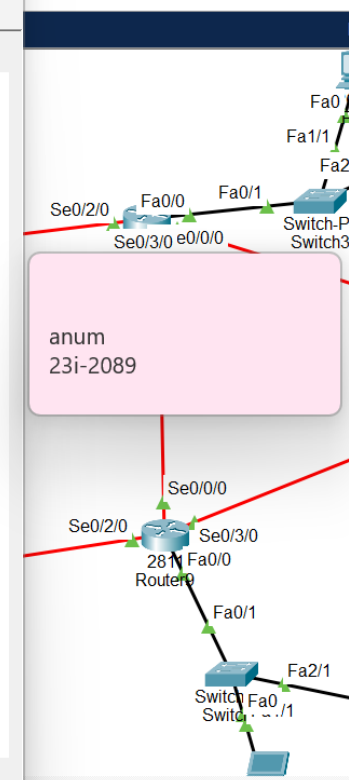
```
Router>en
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status
FastEthernet0/0	120.145.79.33	YES	manual	up
FastEthernet0/1	unassigned	YES	unset	administratively down
Serial0/0/0	120.145.78.17	YES	manual	up
Serial0/1/0	unassigned	YES	unset	administratively down
Serial0/2/0	unassigned	YES	manual	administratively down
Serial0/3/0	120.145.78.14	YES	manual	up
Vlan1	unassigned	YES	unset	administratively down

Router#

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1.2. Correct Routing Protocol on OSPF, EIGRP and RIP

A. On routers of Area1/ OSPF1.

Router0:

The screenshot displays a network simulator interface. The main window is titled 'Router0' and has tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, showing the 'IOS Command Line Interface'. The prompt is 'Router>' and the user has entered the command 'show run | section router ospf'. The output shows the OSPF configuration for Router0:

```
Router>en
Router#show run | section router ospf
router ospf 1
router-id 1.1.1.2
log-adjacency-changes
network 120.145.77.0 0.0.0.3 area 1
network 120.145.0.0 0.0.0.255 area 1
network 120.145.76.0 0.0.0.255 area 1
network 120.145.77.0 0.0.0.255 area 1
Router#
```

A pink tooltip is visible over the CLI window, displaying the text 'lanum 23i-2089'. To the right of the CLI window, a network diagram is partially visible, showing a connection between 'PC-PT PC3' and 'Router5' via interfaces 'Fa0' and 'Fa1'. The diagram also shows a connection between 'Router5' and another device via 'Fa0/1'.

On Router1:

The screenshot displays a network simulator interface. The main window is titled "Router1" and has tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, showing the "IOS Command Line Interface". The prompt is "Router>". The command "show run | section router ospf" has been entered, and the output is displayed below:

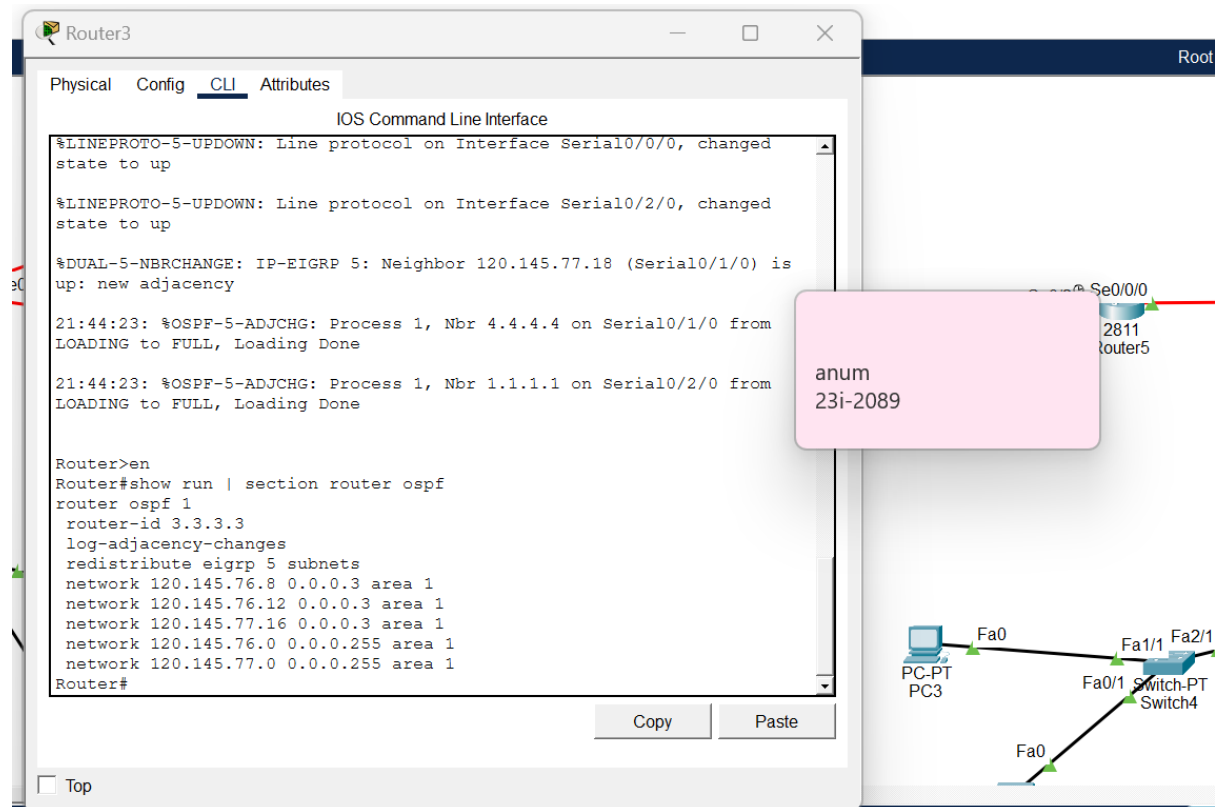
```
Router>en
Router#show run | section router ospf
router ospf 1
  router-id 1.1.1.1
  log-adjacency-changes
  network 120.145.76.0 0.0.0.255 area 1
  network 120.145.77.0 0.0.0.3 area 1
  network 120.145.77.4 0.0.0.3 area 1
  network 120.145.77.8 0.0.0.3 area 1
  network 120.145.76.32 0.0.0.31 area 1
  network 120.145.77.0 0.0.0.255 area 1
  network 120.145.128.0 0.0.0.255 area 1
  network 120.146.0.0 0.0.0.255 area 1
Router#
```

Below the CLI window, there are "Copy" and "Paste" buttons. To the right of the CLI window, there is a network diagram. It shows a PC labeled "PC-PT PC3" connected to a switch labeled "Fa0". The switch is also connected to a router labeled "Fa0/1". The router is connected to another switch labeled "Fa0". The switch is connected to a PC labeled "PC-PT PC3".

A pink tooltip is visible over the CLI window, containing the text:

```
panum
23i-2089
```

On Router3:



The screenshot shows the Router3 CLI window with the following output:

```
Router3
Physical Config CLI Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed
state to up
%DUAL-5-NBRCHANGE: IP-EIGRP 5: Neighbor 120.145.77.18 (Serial0/1/0) is
up: new adjacency
21:44:23: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Serial0/1/0 from
LOADING to FULL, Loading Done
21:44:23: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial0/2/0 from
LOADING to FULL, Loading Done

Router>en
Router#show run | section router ospf
router ospf 1
router-id 3.3.3.3
log-adjacency-changes
redistribute eigrp 5 subnets
network 120.145.76.8 0.0.0.3 area 1
network 120.145.76.12 0.0.0.3 area 1
network 120.145.77.16 0.0.0.3 area 1
network 120.145.76.0 0.0.0.255 area 1
network 120.145.77.0 0.0.0.255 area 1
Router#
```

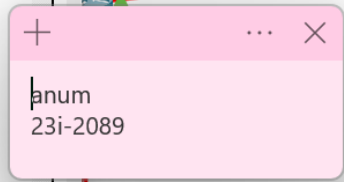
A pink callout box contains the text:

```
anum
23i-2089
```

The network diagram shows a topology with the following components and connections:

- PC-PT PC3 connected to Fa0 of a switch.
- The switch connected to Fa0/1 of Switch-PT Switch4.
- Switch-PT Switch4 connected to Fa0 of another switch.
- The second switch connected to Fa1/1 and Fa2/1 of Router5.
- Router5 connected to Se0/0/0 of Router3.

On Router4:



Router7:

The screenshot shows the Router7 CLI interface with the following content:

Physical Config **CLI** Attributes

IOS Command Line Interface

%DUAL-5-NBRCHANGE: IP-EIGRP 5: Neighbor 120.145.78.26 (Serial0/3/0) is up: new adjacency

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 5: Neighbor 120.145.78.9 (Serial0/2/0) is up: new adjacency

Router>en

Router#show run | section router eigrp

router eigrp 5

network 120.145.78.0 0.0.0.31

network 120.145.79.0 0.0.0.127

network 120.145.78.0 0.0.0.255

network 120.145.79.0 0.0.0.31

network 120.145.78.8 0.0.0.3

network 120.145.78.0 0.0.0.63

Router#show ip eigrp neighbors

IP-EIGRP neighbors for process 5

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
0	120.145.78.22	Se0/0/0	10	00:22:31	40	1000	0	80
1	120.145.78.26	Se0/3/0	13	00:22:28	40	1000	0	93
2	120.145.78.9	Se0/2/0	12	00:22:28	40	1000	0	89

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Diagram annotations: Red arrows point from the CLI output to a network diagram. One arrow points from 'Se0/3/0' to a box labeled 'anum 23i-2089'. Another arrow points from 'Se0/2/0' to a box labeled 'Se0/2/c'.

C. On Router16 for RIP block

 Router16

Physical Config CLI Attributes

IOS Co

```
Router16>en
Router#show run | section router rip
router rip
  version 2
  network 120.0.0.0
  no auto-summary
Router#show ip rip database
120.145.0.0/24      auto-summary
120.145.0.0/24
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.76.12/30   auto-summary
120.145.76.12/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.77.0/30    auto-summary
120.145.77.0/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.77.4/30    auto-summary
120.145.77.4/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.77.8/30    auto-summary
120.145.77.8/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.77.16/30   auto-summary
120.145.77.16/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.78.0/30    auto-summary
120.145.78.0/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.78.4/30    auto-summary
120.145.78.4/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.78.8/30    auto-summary
120.145.78.8/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.78.12/30   auto-summary
120.145.78.12/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
120.145.78.16/30   auto-summary
120.145.78.16/30
  [2] via 120.145.82.41, 00:00:04, Serial0/1/0
```

+

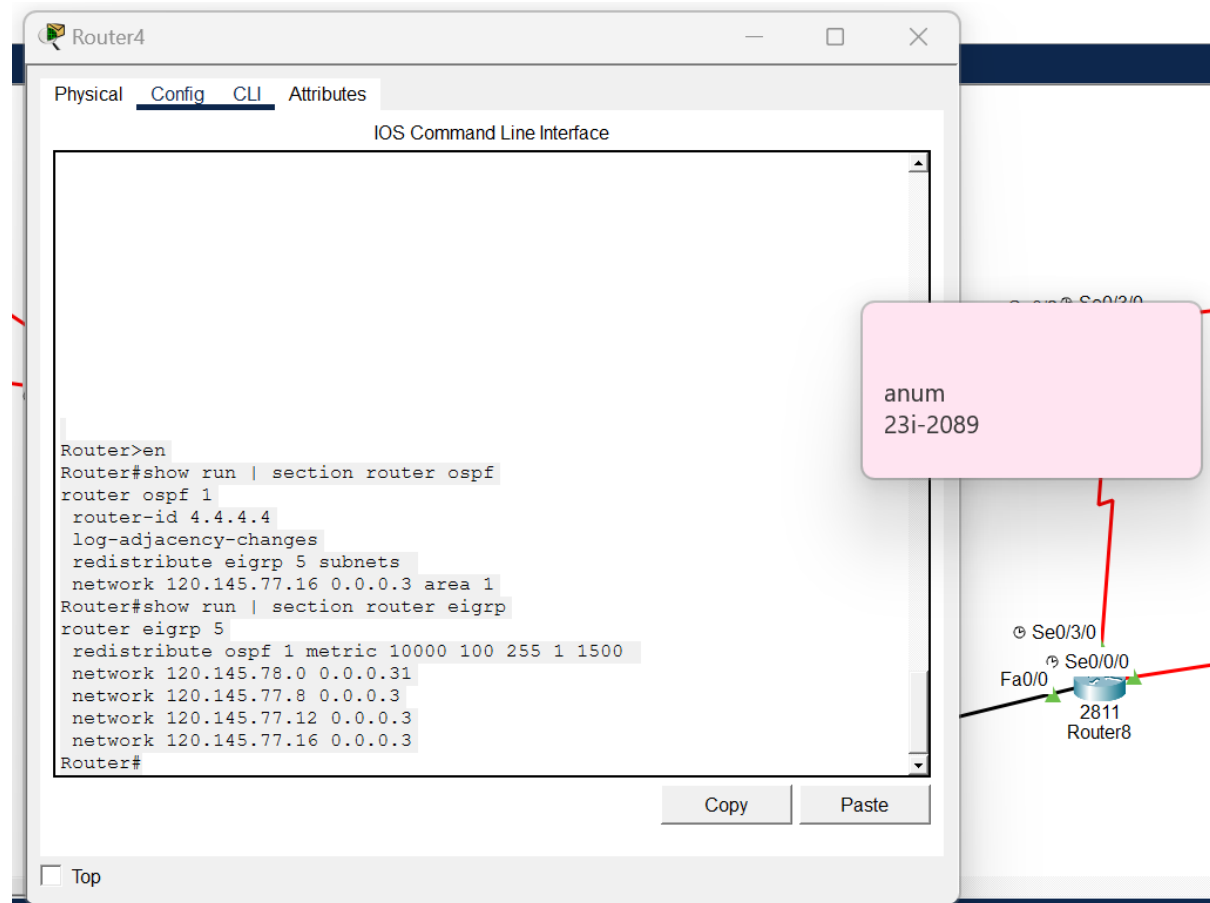
...

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2. Redistribution Of Routing Between Areas (OSPF -> EIGRP5 -> RIP)

2.1. On Router4, that is the router between OSPF1 and EIGRP5

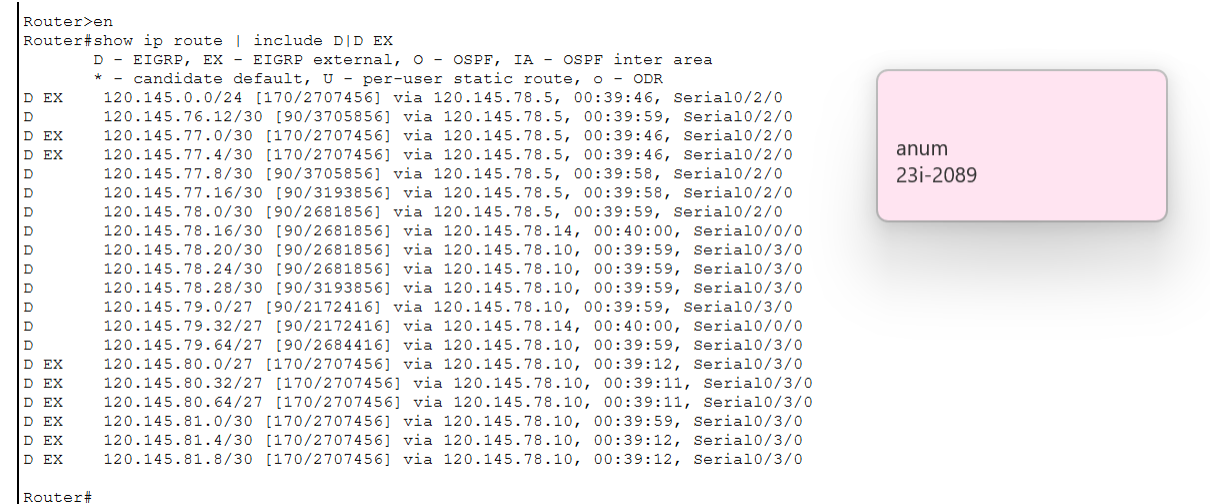


The screenshot shows the configuration window for Router4. The 'Config' tab is active, displaying the 'IOS Command Line Interface'. The configuration includes OSPF and EIGRP settings. A pink callout box points to the 'Se0/3/0' interface of Router4, which is connected to the 'Se0/0/0' interface of Router8 (2811). The callout box contains the text 'anum 23i-2089'.

```
Router>en
Router#show run | section router ospf
router ospf 1
  router-id 4.4.4.4
  log-adjacency-changes
  redistribute eigrp 5 subnets
  network 120.145.77.16 0.0.0.3 area 1
Router#show run | section router eigrp
router eigrp 5
  redistribute ospf 1 metric 10000 100 255 1 1500
  network 120.145.78.0 0.0.0.31
  network 120.145.77.8 0.0.0.3
  network 120.145.77.12 0.0.0.3
  network 120.145.77.16 0.0.0.3
Router#
```

2.2. At Router6 and Router7, EIGRP routes + D EX routes for OSPF networks (like 120.146.0.0/24) is displayed.

Router6:



The screenshot shows the configuration window for Router6. The 'Config' tab is active, displaying the 'IOS Command Line Interface'. The configuration includes EIGRP and D EX routes for OSPF networks. A pink callout box points to the 'Se0/3/0' interface of Router6, which is connected to the 'Se0/0/0' interface of Router8 (2811). The callout box contains the text 'anum 23i-2089'.

```
Router>en
Router#show ip route | include D EX
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
* - candidate default, U - per-user static route, o - ODR
D EX 120.145.0.0/24 [170/2707456] via 120.145.78.5, 00:39:46, Serial0/2/0
D 120.145.76.12/30 [90/3705856] via 120.145.78.5, 00:39:59, Serial0/2/0
D EX 120.145.77.0/30 [170/2707456] via 120.145.78.5, 00:39:46, Serial0/2/0
D EX 120.145.77.4/30 [170/2707456] via 120.145.78.5, 00:39:46, Serial0/2/0
D 120.145.77.8/30 [90/3705856] via 120.145.78.5, 00:39:58, Serial0/2/0
D 120.145.77.16/30 [90/3193856] via 120.145.78.5, 00:39:58, Serial0/2/0
D 120.145.78.0/30 [90/2681856] via 120.145.78.5, 00:39:59, Serial0/2/0
D 120.145.78.16/30 [90/2681856] via 120.145.78.14, 00:40:00, Serial0/0/0
D 120.145.78.20/30 [90/2681856] via 120.145.78.10, 00:39:59, Serial0/3/0
D 120.145.78.24/30 [90/2681856] via 120.145.78.10, 00:39:59, Serial0/3/0
D 120.145.78.28/30 [90/3193856] via 120.145.78.10, 00:39:59, Serial0/3/0
D 120.145.79.0/27 [90/2172416] via 120.145.78.10, 00:39:59, Serial0/3/0
D 120.145.79.32/27 [90/2172416] via 120.145.78.14, 00:40:00, Serial0/0/0
D 120.145.79.64/27 [90/2684416] via 120.145.78.10, 00:39:59, Serial0/3/0
D EX 120.145.80.0/27 [170/2707456] via 120.145.78.10, 00:39:12, Serial0/3/0
D EX 120.145.80.32/27 [170/2707456] via 120.145.78.10, 00:39:11, Serial0/3/0
D EX 120.145.80.64/27 [170/2707456] via 120.145.78.10, 00:39:11, Serial0/3/0
D EX 120.145.81.0/30 [170/2707456] via 120.145.78.10, 00:39:59, Serial0/3/0
D EX 120.145.81.4/30 [170/2707456] via 120.145.78.10, 00:39:12, Serial0/3/0
D EX 120.145.81.8/30 [170/2707456] via 120.145.78.10, 00:39:12, Serial0/3/0
Router#
```

Router7:

```
Router>en
Router#show ip route | include D\|D EX
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
* - candidate default, U - per-user static route, o - ODR
D EX 120.145.0.0/24 [170/3219456] via 120.145.78.13, 00:58:14, Serial0/3/0
D 120.145.76.12/30 [90/4217856] via 120.145.78.13, 00:58:27, Serial0/3/0
D EX 120.145.77.0/30 [170/3219456] via 120.145.78.13, 00:58:14, Serial0/3/0
D EX 120.145.77.4/30 [170/3219456] via 120.145.78.13, 00:58:14, Serial0/3/0
D 120.145.77.8/30 [90/4217856] via 120.145.78.13, 00:58:26, Serial0/3/0
D 120.145.77.16/30 [90/3705856] via 120.145.78.13, 00:58:26, Serial0/3/0
D 120.145.78.0/30 [90/3193856] via 120.145.78.13, 00:58:27, Serial0/3/0
D 120.145.78.4/30 [90/2681856] via 120.145.78.13, 00:58:27, Serial0/3/0
D 120.145.78.8/30 [90/2681856] via 120.145.78.13, 00:58:28, Serial0/3/0
D 120.145.78.20/30 [90/2681856] via 120.145.78.18, 00:58:27, Serial0/0/0
D 120.145.78.24/30 [90/3193856] via 120.145.78.13, 00:58:27, Serial0/3/0
D 120.145.78.28/30 [90/2681856] via 120.145.78.18, 00:58:27, Serial0/0/0
D 120.145.79.0/27 [90/2684416] via 120.145.78.13, 00:58:27, Serial0/3/0
D 120.145.79.64/27 [90/2172416] via 120.145.78.18, 00:58:27, Serial0/0/0
D EX 120.145.80.0/27 [170/2707456] via 120.145.78.18, 00:57:39, Serial0/0/0
D EX 120.145.80.32/27 [170/2707456] via 120.145.78.18, 00:57:39, Serial0/0/0
D EX 120.145.80.64/27 [170/2707456] via 120.145.78.18, 00:57:39, Serial0/0/0
D EX 120.145.81.0/30 [170/2707456] via 120.145.78.18, 00:58:27, Serial0/0/0
D EX 120.145.81.4/30 [170/2707456] via 120.145.78.18, 00:57:40, Serial0/0/0
D EX 120.145.81.8/30 [170/2707456] via 120.145.78.18, 00:57:39, Serial0/0/0

Router#
```

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2.3. At router16 for Rip area

The screenshot shows the Router16 CLI interface with the 'CLI' tab selected. The command 'show ip route 120.146.0.0' has been executed, displaying the following output:

```
Router>en
Router#show ip route 120.146.0.0
Routing entry for 120.146.0.0/24
Known via "rip", distance 120, metric 2
  Redistributing via rip
  Last update from 120.145.82.41 on Serial0/1/0, 00:00:25 ago
  Routing Descriptor Blocks:
    * 120.145.82.41, from 120.145.82.41, 00:00:25 ago, via Serial0/1/0
      Route metric is 2, traffic share count is 1

Router#
```

A pink callout box with a plus icon and a close icon contains the text 'anum 23i-2089'. In the background, a network diagram is visible, showing a router labeled 'Router16' connected to a network with interfaces 'Se0/1/0' and 'Se0/0/0'.

3. DHCP in OSPF, EIGRP and RIP

DHCP pools in server3 (RIP) and server0 (OSPF1)

Server0:

The screenshot shows the 'Server0' configuration window with the 'Services' tab selected. On the left, a list of services includes HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP, and PRP. The 'DHCP' service is selected, and its configuration is shown on the right. The 'Interface' is set to 'FastEthernet0' and the 'Service' is 'On'. The 'Pool Name' is 'serverPool'. The 'Default Gateway' is '0.0.0.0'. The 'DNS Server' is '0.0.0.0'. The 'Start IP Address' is '120.146.0.0' and the 'Subnet Mask' is '255.255.255.0'. The 'Maximum Number of Users' is '512'. The 'TFTP Server' is '0.0.0.0' and the 'WLC Address' is '0.0.0.0'. Below the configuration fields are 'Add', 'Save', and 'Remove' buttons. At the bottom, a table lists the configured DHCP pools.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
EIGRP_D_...	120.1...	120.1...	120.1...	255.2...	30	0.0.0.0	0.0.0.0
AREA1_L...	120.1...	120.1...	120.1...	255.2...	50	0.0.0.0	0.0.0.0
AREA1_P...	120.1...	120.1...	120.1...	255.2...	50	0.0.0.0	0.0.0.0
AREA1_N...	120.1...	120.1...	120.1...	255.2...	30	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	120.1...	255.2...	512	0.0.0.0	0.0.0.0

Server3:

Interface: FastEthernet0 Service: On

Pool Name: serverPool

Default Gateway: 10.10.10.1

DNS Server: 8.8.8.8

Start IP Address: 10.10.10.0

Subnet Mask: 255.255.255.0

Maximum Number of Users: 10

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	10.10.10.1	8.8.8.8	10.10.10.0	255.255.255.0	10	0.0.0.0	0.0.0.0
R16_LAN	120.145.0.1	8.8.8.8	120.145.0.20	255.255.255.0	30	0.0.0.0	0.0.0.0
R18_LAN	120.145.0.1	8.8.8.8	120.145.0.29	255.255.255.0	29	0.0.0.0	0.0.0.0

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3.1. at laptop1 (network A) of OSPF

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address...: FE80::20C:CFE:FEA3:D906
IPv6 Address...: ::
IPv4 Address...: 120.145.0.20
Subnet Mask...: 255.255.255.0
Default Gateway...: 120.145.0.1

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: 0.0.0.0

C:\>ping 120.145.0.1

Pinging 120.145.0.1 with 32 bytes of data:

Reply from 120.145.0.1: bytes=32 time=10ms TTL=255
Reply from 120.145.0.1: bytes=32 time=1ms TTL=255
Reply from 120.145.0.1: bytes=32 time=2ms TTL=255

Ping statistics for 120.145.0.1:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 4ms
```

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```
C:\>ping 120.146.0.20

Pinging 120.146.0.20 with 32 bytes of data:

Reply from 120.146.0.20: bytes=32 time=1ms TTL=126
Reply from 120.146.0.20: bytes=32 time=2ms TTL=126
Reply from 120.146.0.20: bytes=32 time=10ms TTL=126
Reply from 120.146.0.20: bytes=32 time=2ms TTL=126

Ping statistics for 120.146.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 3ms

C:\>
```

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3.2. At Network B of OSPF.

Laptop3:

Laptop3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::204:9AFF:FE29:75DD
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 120.145.128.10
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                120.145.128.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 120.145.128.1

Pinging 120.145.128.1 with 32 bytes of data:

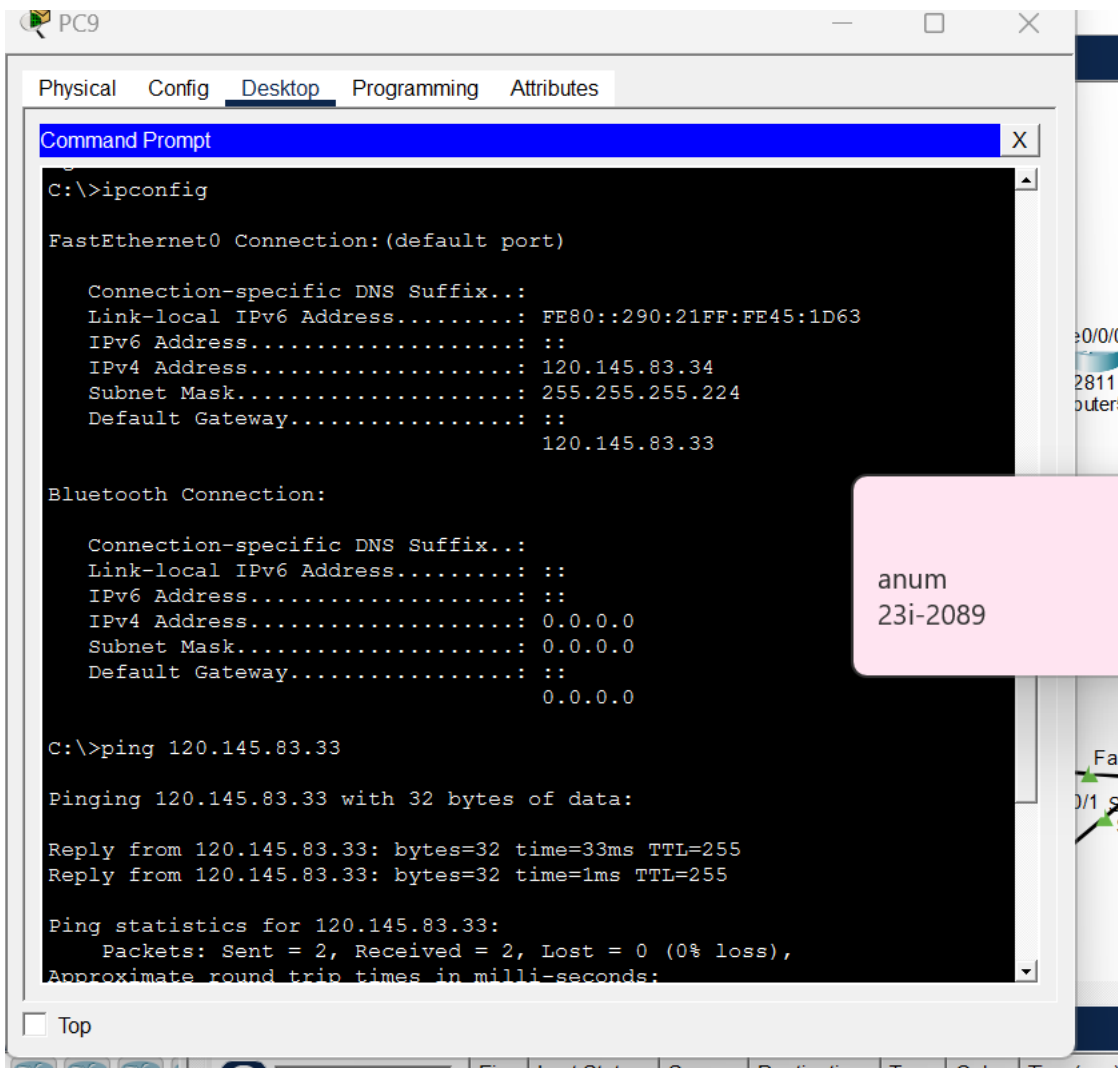
Reply from 120.145.128.1: bytes=32 time<1ms TTL=255
Reply from 120.145.128.1: bytes=32 time<1ms TTL=255
Reply from 120.145.128.1: bytes=32 time=1ms TTL=255

Ping statistics for 120.145.128.1:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

+ ... X
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3.3. At Network C OSPF1

3.4. Network D in EIGRP5 Block



```

C:\>ping 120.145.83.33

Pinging 120.145.83.33 with 32 bytes of data:

Reply from 120.145.83.33: bytes=32 time=33ms TTL=255
Reply from 120.145.83.33: bytes=32 time=1ms TTL=255

Ping statistics for 120.145.83.33:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 33ms, Average = 17ms

Control-C
^C
C:\>ping 120.145.82.41

Pinging 120.145.82.41 with 32 bytes of data:

Reply from 120.145.82.41: bytes=32 time=4ms TTL=254
Reply from 120.145.82.41: bytes=32 time=1ms TTL=254
Reply from 120.145.82.41: bytes=32 time=1ms TTL=254
Reply from 120.145.82.41: bytes=32 time=1ms TTL=254

Ping statistics for 120.145.82.41:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 1ms

```

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3.5. Network J in RIP

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:96FF:FEB5:1308
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 120.145.83.35
    Subnet Mask . . . . .: 255.255.255.224
    Default Gateway . . . . .: ::
                                120.145.83.33

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 120.145.83.33

Pinging 120.145.83.33 with 32 bytes of data:

Reply from 120.145.83.33: bytes=32 time=10ms TTL=255
Reply from 120.145.83.33: bytes=32 time<1ms TTL=255
Reply from 120.145.83.33: bytes=32 time<1ms TTL=255
Reply from 120.145.83.33: bytes=32 time<1ms TTL=255

Ping statistics for 120.145.83.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

```

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4. NAT Verification

4.1. Configuration and Mapping

The screenshot displays the Cisco Packet Tracer interface. A window titled "Router16" is open, showing the "CLI" (Command Line Interface) tab. The command history shows the following commands:

```
Router>en
Router#enable
Router#show run | include ip nat
ip nat inside
ip nat inside
ip nat inside
ip nat outside
ip nat inside source static 120.145.76.181 237.156.89.249
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status
FastEthernet0/0	120.145.83.33	YES	manual	up
FastEthernet0/1	120.145.76.161	YES	manual	down
Serial0/0/0	120.145.82.26	YES	manual	up
Serial0/1/0	120.145.82.42	YES	manual	up
Serial0/2/0	120.145.82.14	YES	manual	down
Serial0/3/0	120.145.82.22	YES	manual	up
Loopback0	unassigned	YES	unset	up
Vlan1	unassigned	YES	unset	administratively down

```
Router#show ip nat translations
Pro Inside global Inside local Outside local Outside
global
--- 237.156.89.249 120.145.76.181 --- ---
Router#
```

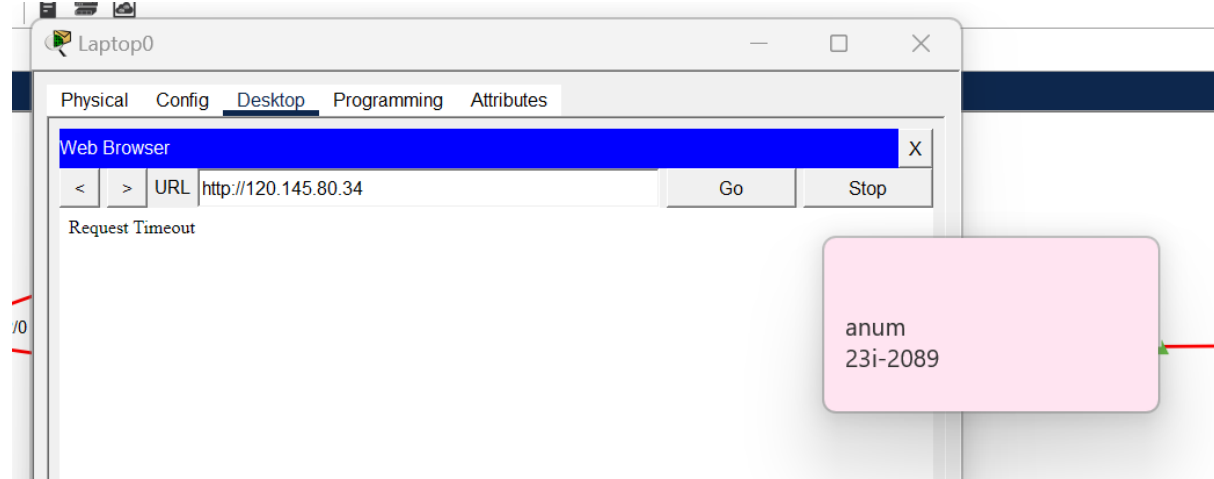
Below the CLI window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

In the background, a network topology is visible. It includes a router labeled "Router5" with a serial interface "Se0/0/0" connected to a red line. Below Router5, there is a switch labeled "Switch-PT" with interfaces "Fa0/1" and "Fa2/1" connected to a green line. A pink callout box with the text "anum 23i-2089" is overlaid on the right side of the image.

5. ACL Behavior and Verification

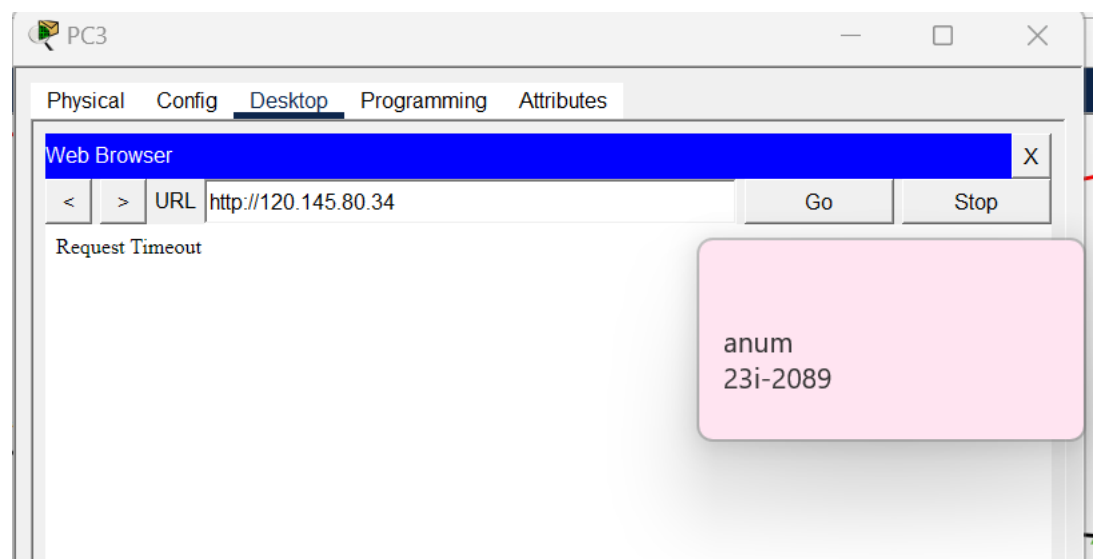
5.1. At network A

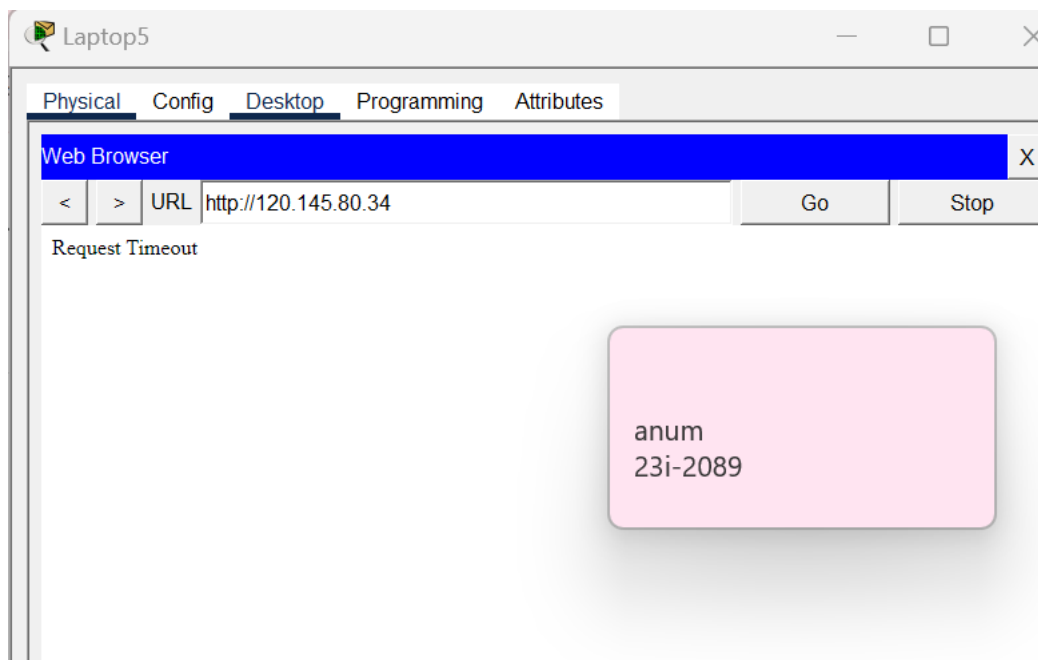
Laptop0 in Network A is blocked from HTTP by using `http://120.145.80.34`.



5.2. At network D

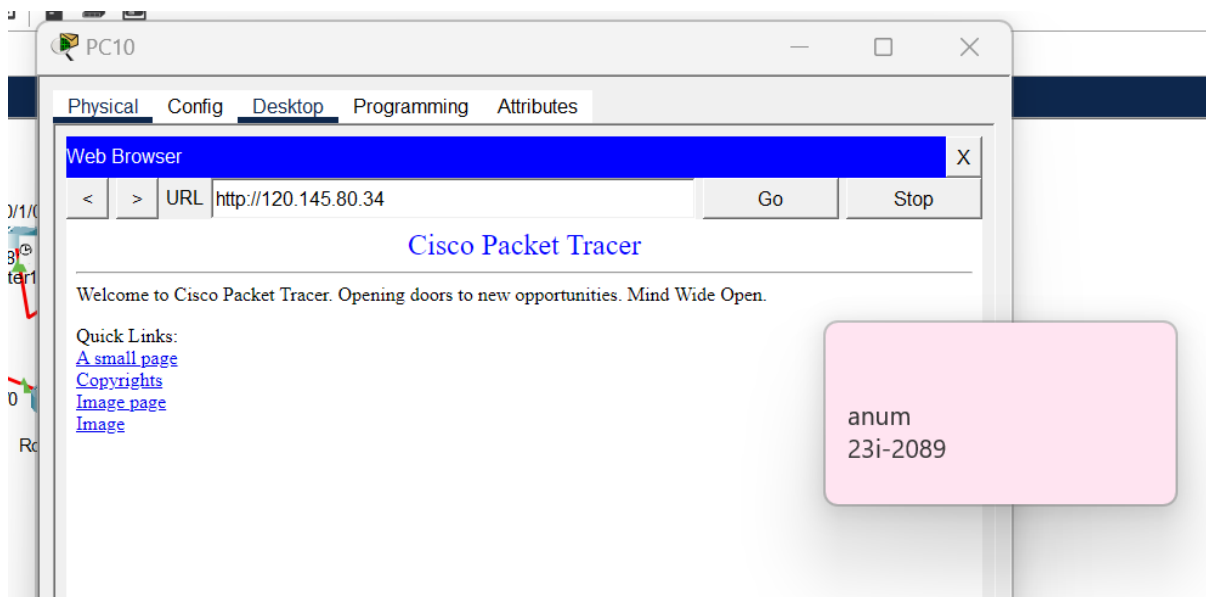
Network D consists of PC3 and Laptop5. Hence the network D is blocked in this case.





5.3. Case of Non-Block

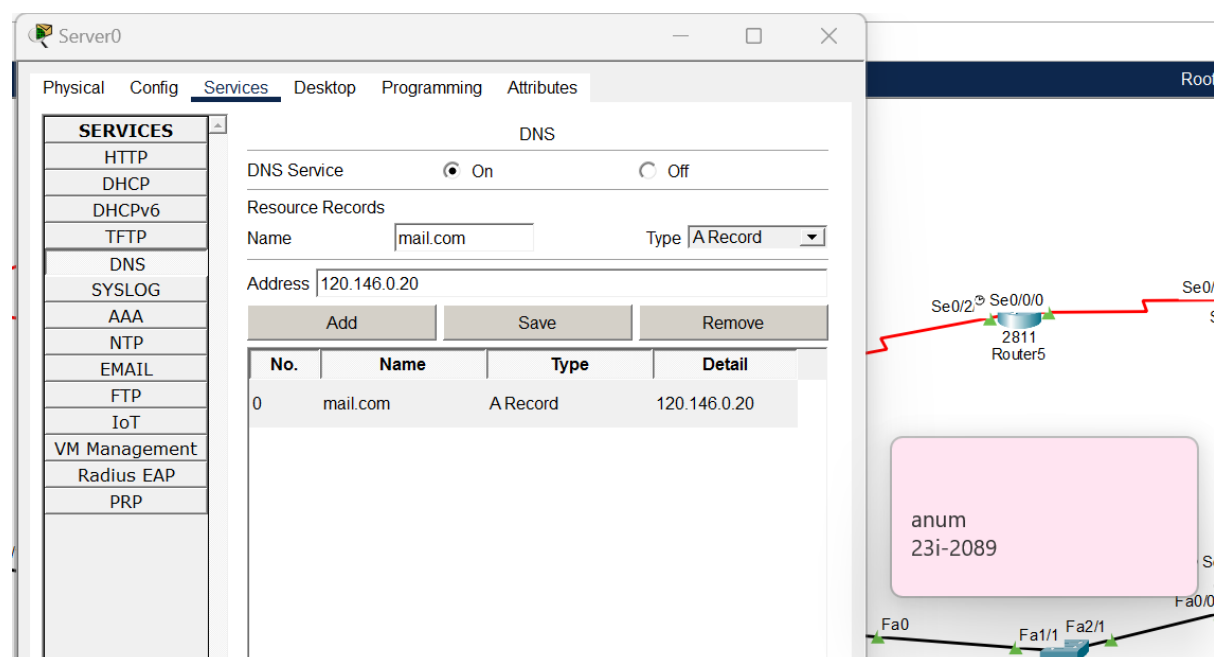
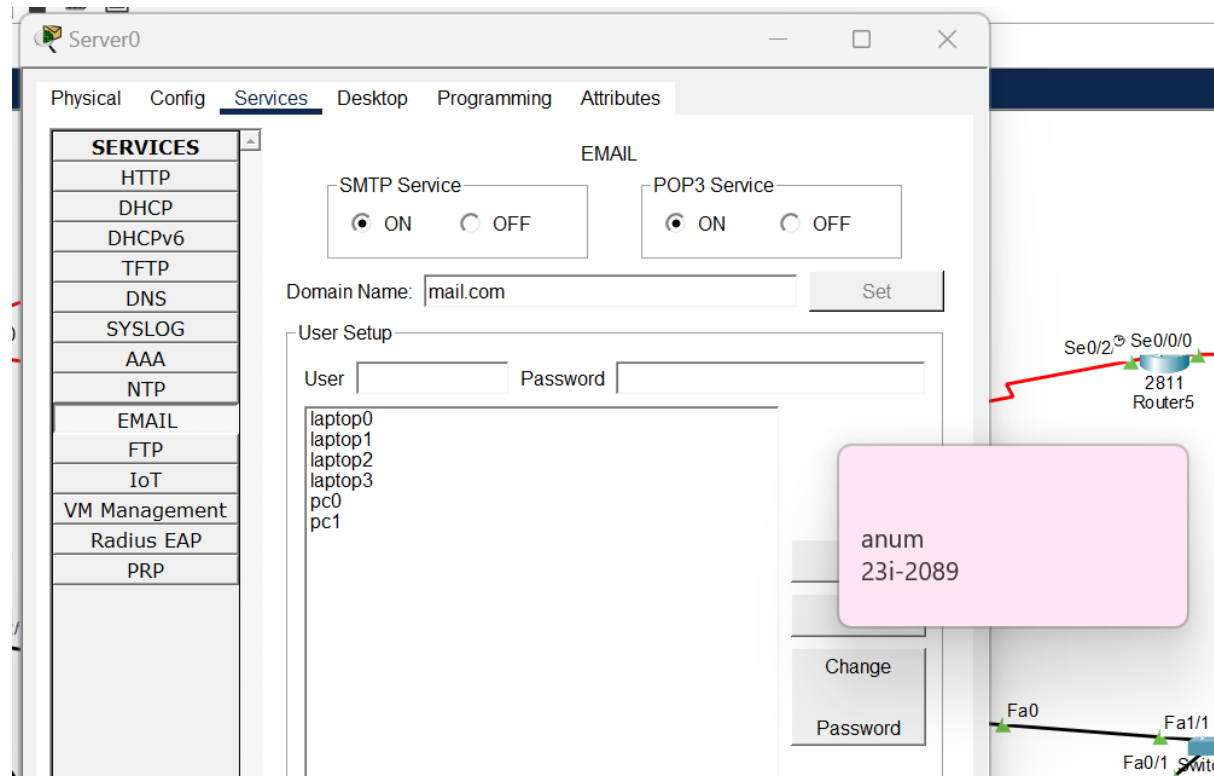
For instance, Pc10 in network J isn't blocked from HTTP. Hence;



6. EMAIL SMTP

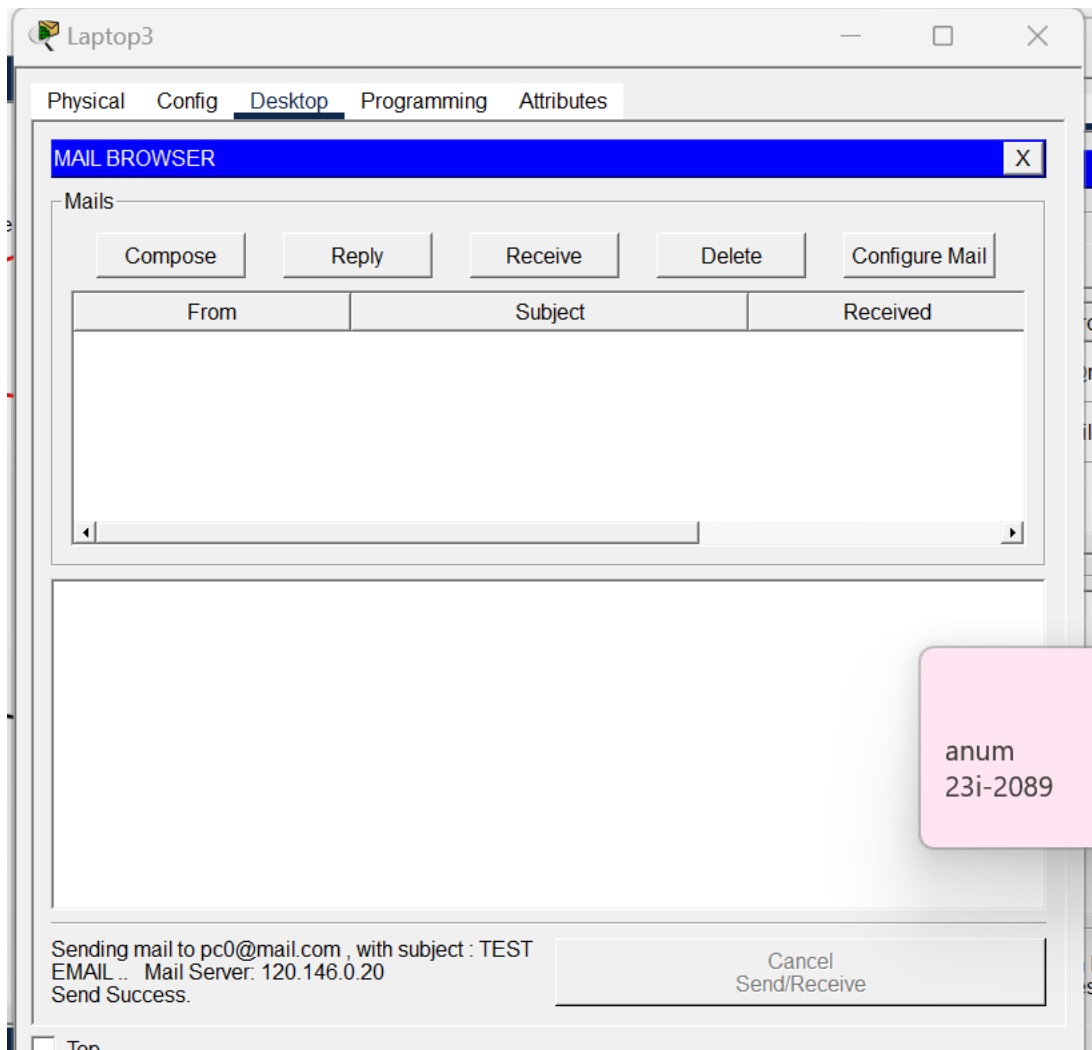
6.1. Confirming Mail Services.

DNS and SMTP were turned on and assigned with domain names, usernames and passwords.

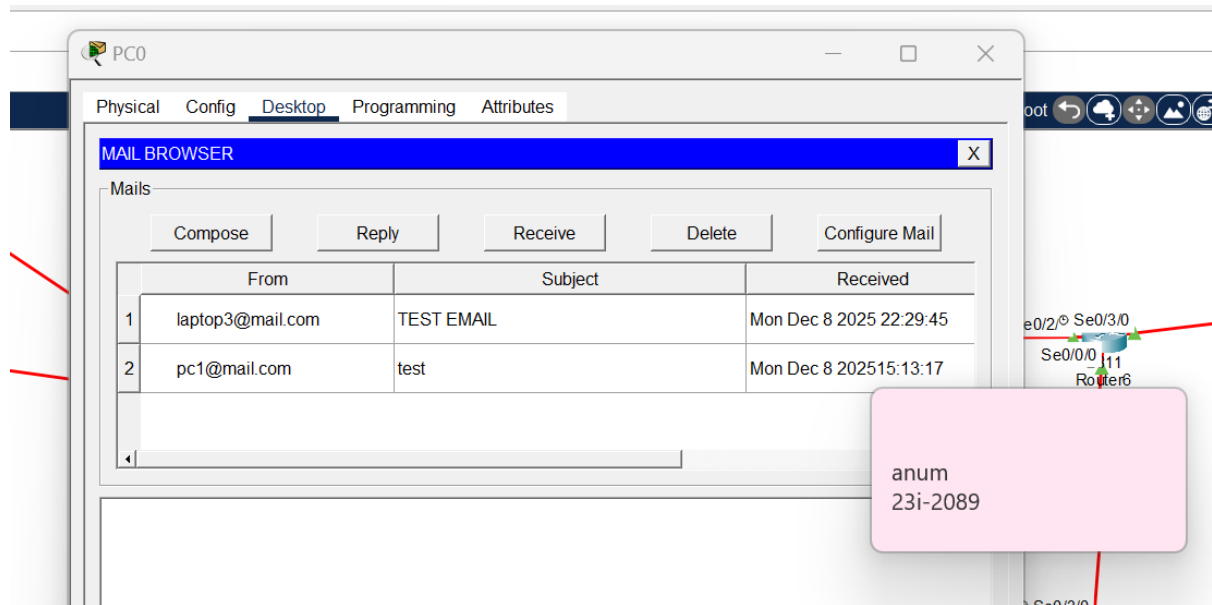


6.2. Testing and Verification

In network A, mail was sent by laptop3 to Pc0.

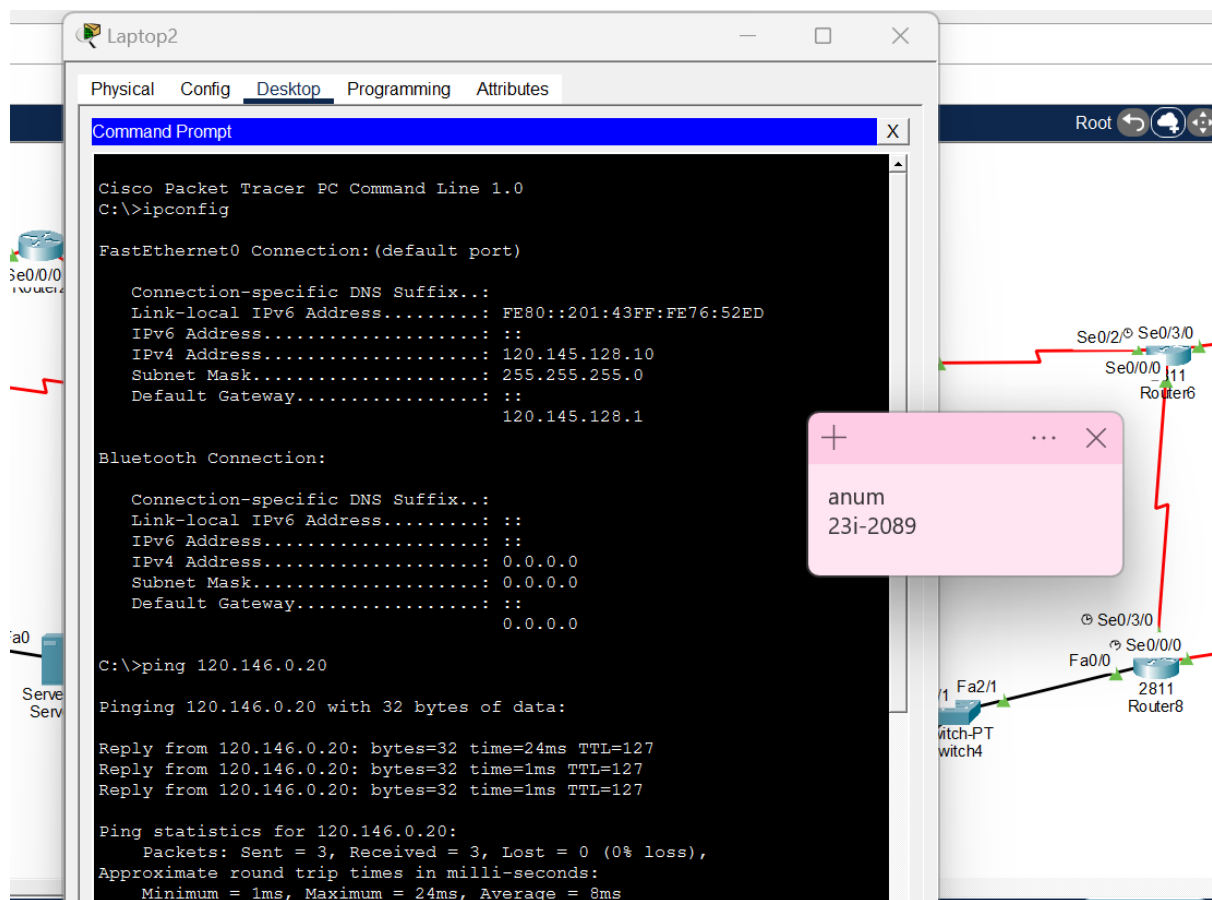


Here, Pc0 has received mail from laptop3.



7. Final Connectivity Check

7.1. For Network B of OSPF1



Laptop6 in Network E obtained IP 120.145.79.66/27 via DHCP with default gateway 120.145.79.65. Successful ping to the gateway confirms local connectivity, and successful ping to 120.146.0.20 did not happen because network E wasn't assigned a DHCP pool.

```
C:\>ping 120.146.0.20

Pinging 120.146.0.20 with 32 bytes of data:

Reply from 120.146.0.20: bytes=32 time=24ms TTL=121
Reply from 120.146.0.20: bytes=32 time=59ms TTL=121

Ping statistics for 120.146.0.20:
    Packets: Sent = 3, Received = 2, Lost = 1 (34% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 24ms, Maximum = 59ms, Average = 41ms

Control-C
^C
C:\>ping 120.145.79.65

Pinging 120.145.79.65 with 32 bytes of data:

Reply from 120.145.79.65: bytes=32 time<1ms TTL=255
Reply from 120.145.79.65: bytes=32 time<1ms TTL=255
Reply from 120.145.79.65: bytes=32 time=1ms TTL=255
Reply from 120.145.79.65: bytes=32 time<1ms TTL=255

Ping statistics for 120.145.79.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



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For connectivity in OSPF2, network G. Hence, Successful ping to the gateway confirms local connectivity.

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:5CFF:FE31:669A
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 120.145.80.3
    Subnet Mask . . . . .: 255.255.255.224
    Default Gateway . . . . .: ::
                                120.145.80.1

Bluetooth Connection:

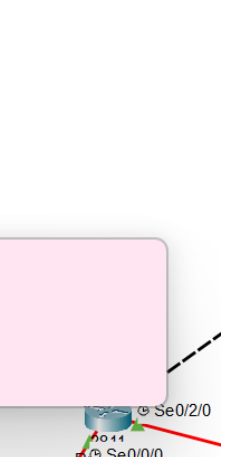
    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 120.145.79.65

Pinging 120.145.79.65 with 32 bytes of data:

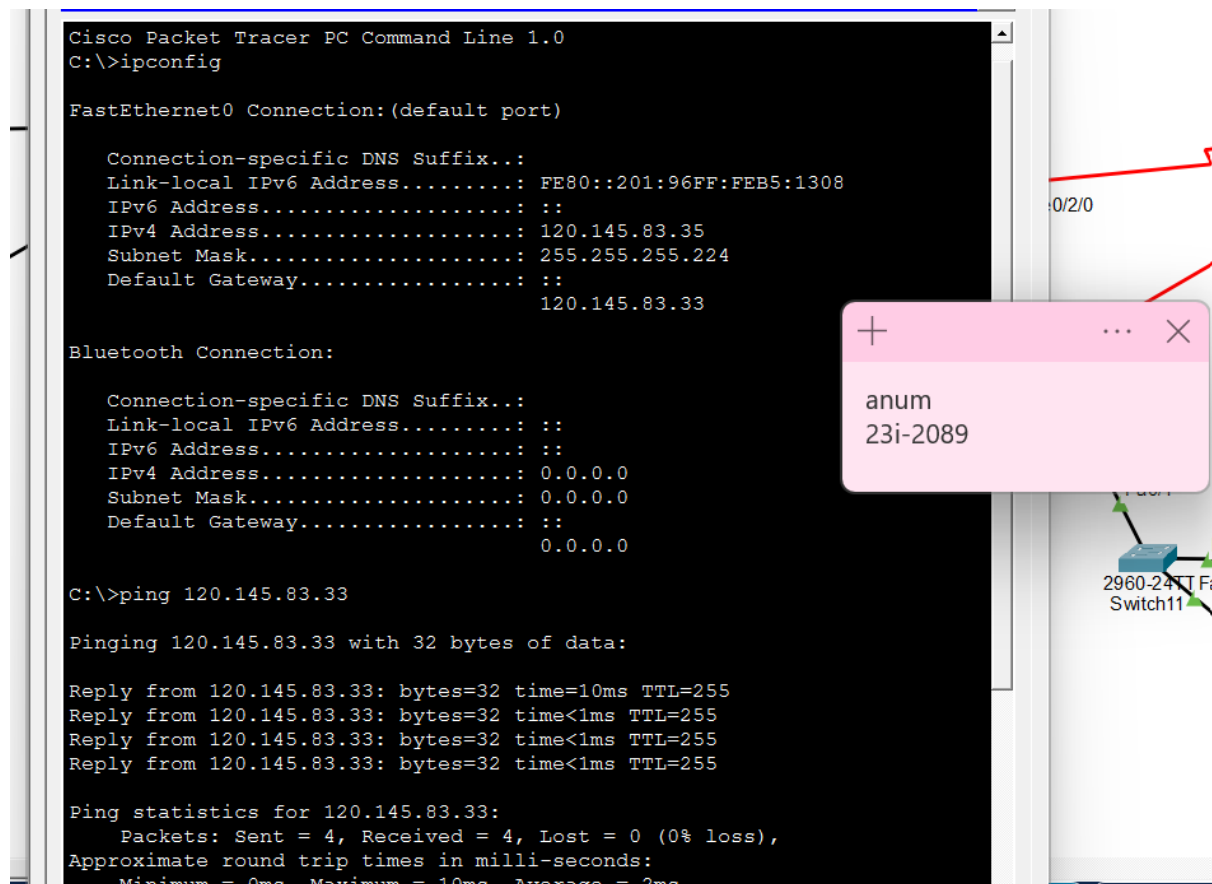
Reply from 120.145.79.65: bytes=32 time=12ms TTL=252
Reply from 120.145.79.65: bytes=32 time=15ms TTL=252
Reply from 120.145.79.65: bytes=32 time=13ms TTL=252

Ping statistics for 120.145.79.65:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 15ms, Average = 13ms
```



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PC10 can reach its gateway. Hence, DHCP and local connectivity are working for Network J.



8. Challenges

- Overlapping in IPs assigned to Networks and in network statements under OSPF/EIGRP.
- Took time to implement redistribution in routers.
- ACL application was causing blockage in both devices of network A.
- Nat was mainly implemented as configuration but couldn't be done as end-to-end for every host.

9. Conclusion

The project was done by using protocols, services and routing. The main areas were OSPF, EIGRP and RIP. The purpose was to interconnect the areas, configuring NAT, selective access to browser via ACL, email in area 1 and DHCP design by creating a topology on Cisco packet tracer.

10. Topology

